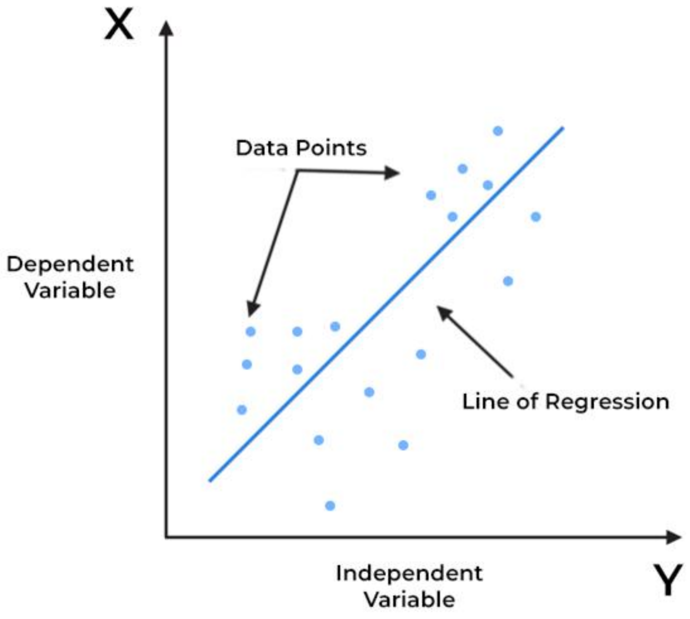
Agenda

- Regression Models
  - Linear Regression
  - Decision Tree Regressor
  - Random Forest Regressor
  - Gradient Boosting Regressor
- Hands on Using Scikit Learn

Different Regression Techniques

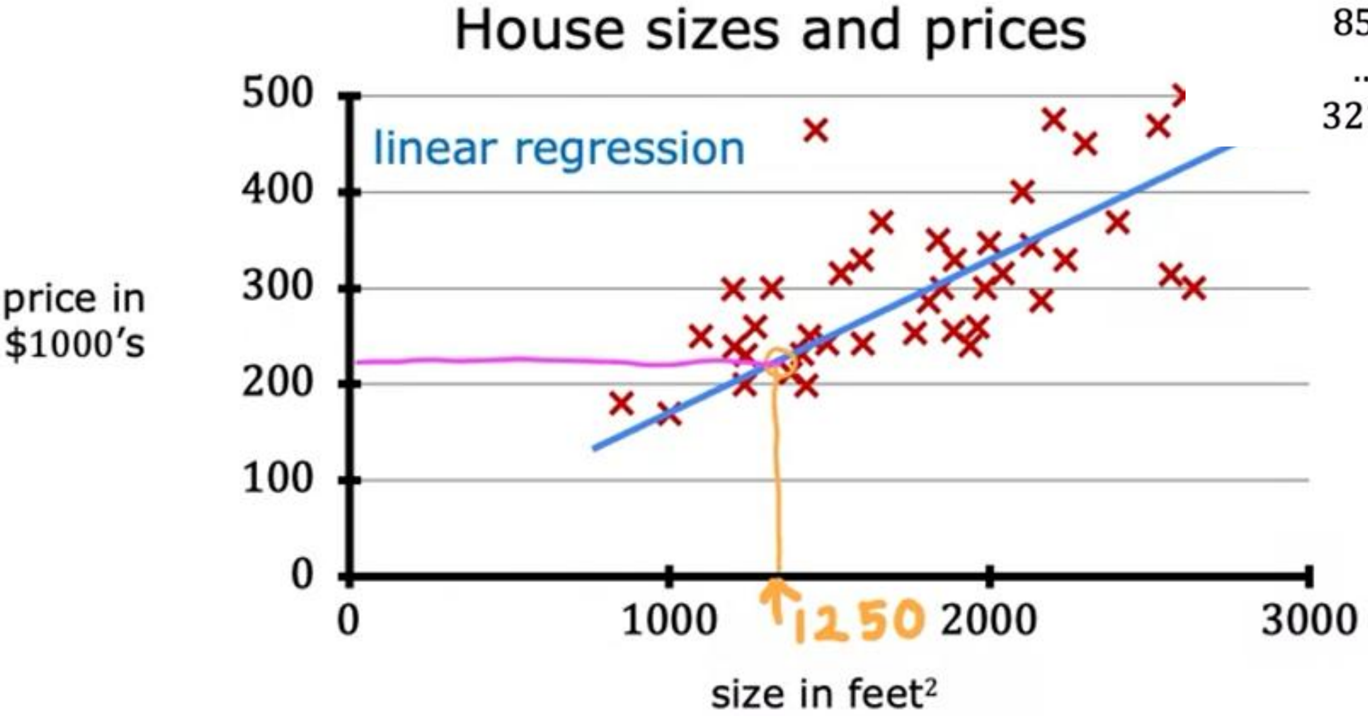
Linear Regression

Linear regression is a **supervised** machine learning algorithm used for predicting a **continuous output** (**dependent variable**) based on one or more **input features** (**independent variables**).



Linear Regression

Linear Regression

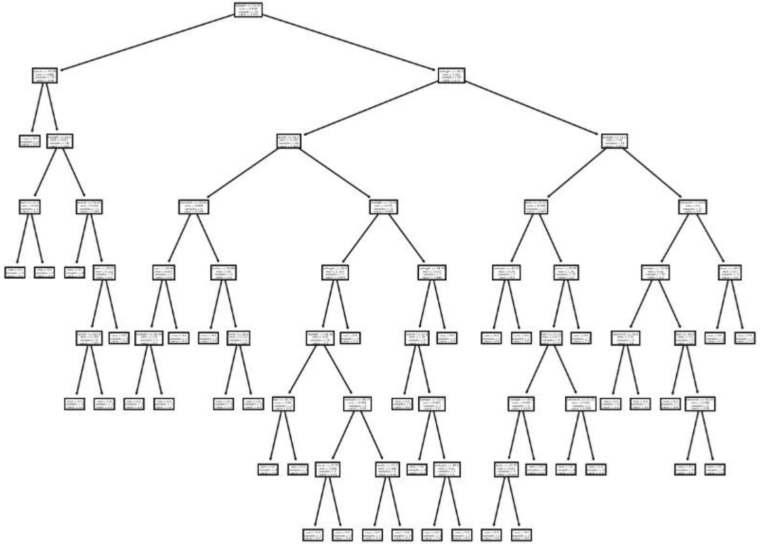


Data table

size in feet²	price in $1000's
2104	400
1416	232
1534	315
852	178
...	...
3210	870

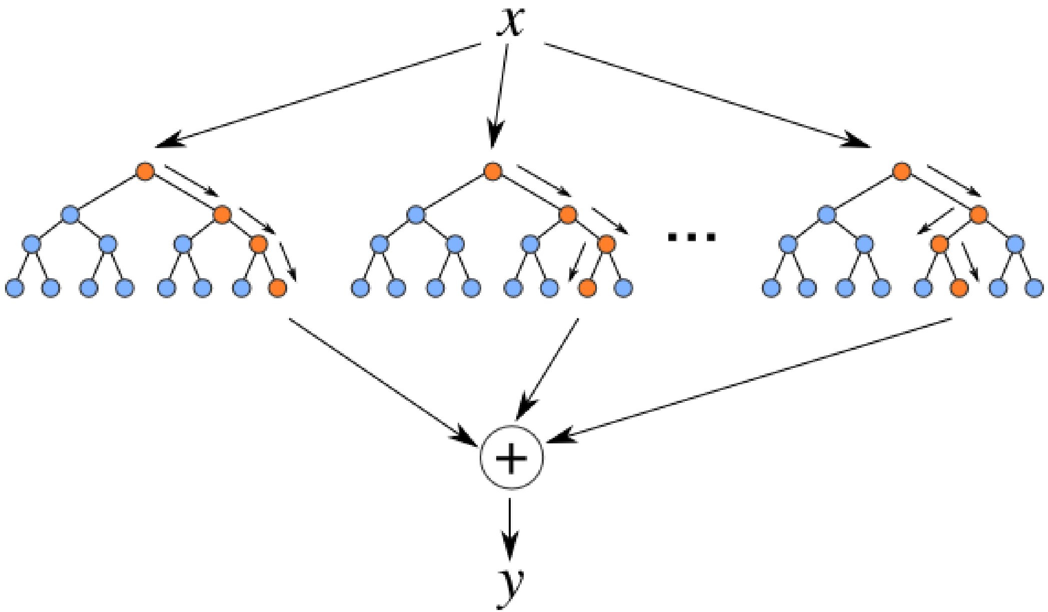
Decision Tree Regressor

Decision tree regression involves **partitioning the data into subsets** based on the values of **independent variables** and predicting the average of the target variable for each subset.



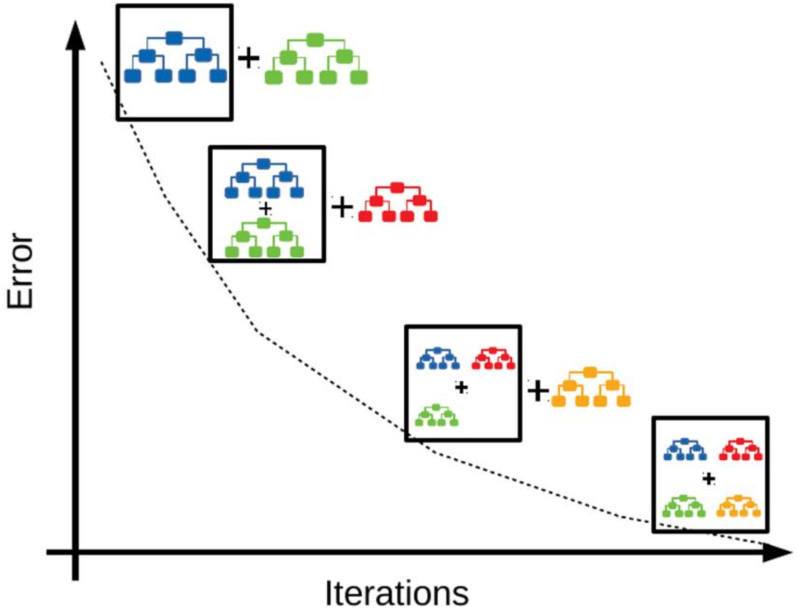
Random Forest Regressor:

Random Forest is learning method that creates **multiple decision trees** during training and outputs the **average prediction** (for regression) from all individual trees.



Gradient Boosting Regressor

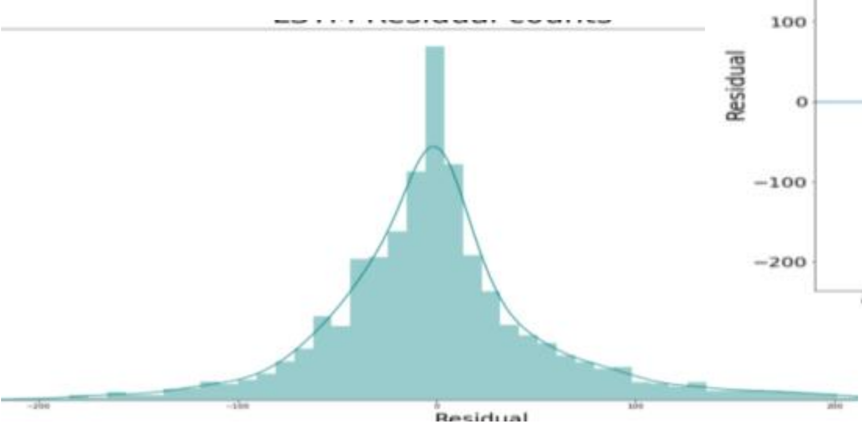
Gradient Boosting is **learning technique** that **builds multiple decision trees** sequentially, each one correcting the errors of its predecessor.

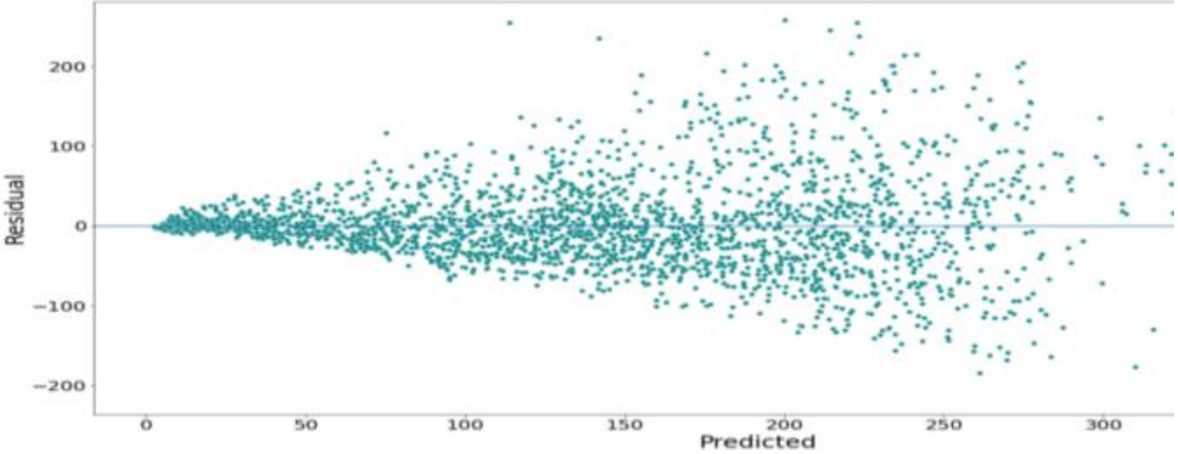


Residual Analysis:

Residual analysis is a **technique for evaluating the validity of a linear regression model by examining the patterns of residuals**.

A residual is the difference between the observed value and the predicted value of a quantity of interest.





Mean Squared Error (MSE)

The **Mean Squared Error (MSE)** is a common evaluation metric used in regression tasks.

It measures the average squared difference between the actual and predicted values.